Problem 1.1. Compute the product of the following two SSYT both ways (insertion and rectification)

$$T = \begin{ytableau} 1 & 2 & 2 & 3 \\ 2 & 3 & 5 & 5 \\ 4 & 4 & 6 \\ 5 & 6 \end{ytableau} \text{ and } U = \begin{ytableau} 1 & 3 \\ 2 \end{ytableau}$$

Proof. **Insertion:** $T \cdot U = ((T \leftarrow 2) \leftarrow 1) \leftarrow 3$

$$(T \leftarrow 2)$$

$$\begin{ytableau} 1 & 2 & 2 & *(red!30) 2 \\ 2 & 3 & 5 & 5 \\ 4 & 4 & 6 \\ 5 & 6 \end{ytableau} \xrightarrow{\curvearrowright 3} \begin{ytableau} 1 & 2 & 2 & 2 \\ 2 & 3 & *(red!30) 3 & 5 \\ 4 & 4 & 6 \\ 5 & 6 \end{ytableau} \xrightarrow{\curvearrowright 5} \begin{ytableau} 1 & 2 & 2 & 2 \\ 2 & 3 & 3 & 5 \\ 4 & 4 & *(red!30) 5 \\ 5 & 6 \end{ytableau} \xrightarrow{\curvearrowright 6} \begin{ytableau} 1 & 2 & 2 & 2 \\ 2 & 3 & 3 & 5 \\ 4 & 4 & 5 \\ 5 & 6 & *(red!30) 6 \end{ytableau}$$

$$(T \leftarrow 2) \leftarrow 1$$

$$\begin{ytableau} 1 & *(red!30) 1 & 2 & 2 \\ 2 & 3 & 3 & 5 \\ 4 & 4 & 5 \\ 5 & 6 & 6 \end{ytableau} \xrightarrow{\curvearrowright 2} \begin{ytableau} 1 & 1 & 2 & 2 \\ 2 & *(red!30) 2 & 3 & 5 \\ 4 & 4 & 5 \\ 5 & 6 & 6 \end{ytableau} \xrightarrow{\curvearrowright 3} \begin{ytableau} 1 & 1 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ *(red!30) 3 & 4 & 5 \\ 5 & 6 & 6 \end{ytableau} \xrightarrow{\curvearrowright 4} \begin{ytableau} 1 & 1 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ 3 & 4 & 5 \\ *(red!30) 4 & 6 & 6 \end{ytableau} \xrightarrow{\curvearrowright 5} \begin{ytableau} 1 & 1 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ 3 & 4 & 5 \\ 4 & 6 & 6 \\ *(red!30) 5 \end{ytableau}$$

$$((T \leftarrow 2) \leftarrow 1) \leftarrow 3 = T \cdot U$$

$$\begin{ytableau} 1 & 1 & 2 & 2 & *(red!30) 3 \\ 2 & 2 & 3 & 5 \\ 3 & 4 & 5 \\ 4 & 6 & 6 \\ 5 \end{ytableau}$$

We do the same product by rectification on the next page.

- **Fulton—Chapter 1:** Exercises 1, 2 (pp. 15–16), compute product both ways
- **Fulton—Chapter 2:** Exercises 1, 2 (pp. 24–26)
- **Fulton—Chapter 3:** Exercises 1-3 (pp. 34–35)
- **Fulton—Chapter 4:** Exercises 12 (pp. 56) algorithm from class, 6,8,10,14 (pp. 52)
- **Fulton—Chapter 5:** Exercises 2 (pp. 66) 2,4,6
- **Fulton—Chapter 6:** Exercises 1,6 (pp. 74-76) 2,4,6 On 6 no need to deduce formula

$$T * U = \begin{ytableau} *(black) & *(black) & *(black) & *(black) & 1 & 3 \\ *(black) & *(black) & *(black) & *(black) & 2 \\ 1 & 2 & 2 & 3 \\ 2 & 3 & 5 & 5 \\ 4 & 4 & 6 \\ 5 & 6 \end{ytableau}$$

We rectify this skew tableau

$$\begin{ytableau} *(black) & *(black) & *(black) & *(black) & 1 & 3 \\ *(black) & *(black) & *(black) & 2 \\ 1 & 2 & 2 & 3 \\ 2 & 3 & 5 & 5 \\ 4 & 4 & 6 \\ 5 & 6 \end{ytableau} \longrightarrow \begin{ytableau} *(black) & *(black) & *(black) & *(black) & 1 & 3 \\ *(black) & *(black) & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 3 & 5 \\ 4 & 4 & 6 \\ 5 & 6 \end{ytableau} \longrightarrow \begin{ytableau} *(black) & *(black) & *(black) & *(black) & 1 & 3 \\ *(black) & 2 & 2 & 2 \\ 1 & 3 & 3 & 5 \\ 2 & 4 & 5 \\ 4 & 6 & 6 \\ 5 \end{ytableau}$$

$$\longrightarrow \begin{ytableau} *(black) & *(black) & *(black) & *(black) & 1 & 3 \\ 1 & 2 & 2 & 2 \\ 2 & 3 & 3 & 5 \\ 4 & 4 & 5 \\ 5 & 6 & 6 \end{ytableau} \longrightarrow \begin{ytableau} *(black) & *(black) & *(black) & 1 & 3 \\ 1 & 2 & 2 & 2 \\ 2 & 3 & 3 & 5 \\ 4 & 4 & 5 \\ 5 & 6 & 6 \end{ytableau} \longrightarrow \begin{ytableau} *(black) & *(black) & 1 & 2 & 3 \\ 1 & 2 & 2 & 5 \\ 2 & 3 & 3 \\ 4 & 4 & 5 \\ 5 & 6 & 6 \end{ytableau}$$

$$\longrightarrow \begin{ytableau} *(black) & 1 & 2 & 2 & 3 \\ 1 & 2 & 3 & 5 \\ 2 & 3 & 5 \\ 4 & 4 & 6 \\ 5 & 6 \end{ytableau} \longrightarrow \begin{ytableau} 1 & 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 5 \\ 3 & 4 & 5 \\ 4 & 6 & 6 \\ 5 \end{ytableau}$$

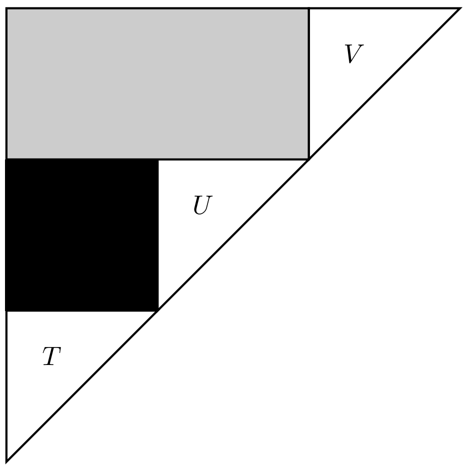
So we see that

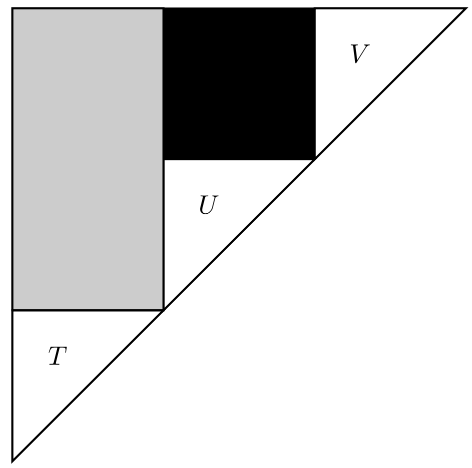
$$\mathrm{Rect}(T * U) = \begin{ytableau} 1 & 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 5 \\ 3 & 4 & 5 \\ 4 & 6 & 6 \\ 5 \end{ytableau} = T \cdot U$$

$\square$

Problem 1.2. Show that associativity of the product (Claim 1) $T \cdot U \cdot V$ follows from Claims 2 and 3 by forming the appropriate skew tableau $T * U * V$ from three given tableaux T, U, V.

Proof. Observe the structure of $T * U * V =$ **[Left]** $(T * U) * V = T * (U * V)$ **[Right]**





Since $T * U * V$ is a skew tableau, by [**Claim 2**], any choice of inner corners lead to the same skew tableau. So starting the *jeu de taquin* by sliding inner corners bounded by U, T and then sliding the remaining inner corners bounded to the right by V will lead to the same rectfication as sliding inner corners bounded by U, V and then sliding the remaining inner corners bounded below by T. That is,

$$\mathrm{Rect}(T * U * V) = \textbf{[Left] } \mathrm{Rect}((\mathrm{Rect}(T * U)) * V) = \mathrm{Rect}(T * (\mathrm{Rect}(U * V))) \textbf{ [Right]}. \quad (i)$$

Well, by **[Claim 3]** the rectfication of $A * B$ is equal to the bump product $A \cdot B$; for any SSYT A, B, $\mathrm{Rect}(A * B) = A \cdot B$. So then in fact (i) implies that

$$T \cdot U \cdot V = (T \cdot U) \cdot V = (T \cdot (U \cdot V)).$$

$\square$

Problem 2.1. Recall the following.

$$S_\lambda S_{(p)} = \sum_\mu S_\mu$$
$$S_\lambda S_{(1^p)} = \sum_\eta S_\eta$$

Where μ are all possible tableau obtained from λ by adding p boxes, no two in the same column, and η are all possible tableau obtained from λ by adding p boxes, no two in the same row.

Prove that if $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 0)$, then the following algebraic conditions hold:

$$\mu_1 \geq \lambda_1 \geq \mu_2 \geq \lambda_2 \geq \cdots \geq \mu_k \geq \lambda_k \geq \mu_{k+1} \geq \mu_{k+2} = 0 \text{ and } \sum \mu_i = \sum \lambda_i + p \tag{1}$$
$$\eta_1 \geq \lambda_1 \geq \eta_2 \geq \lambda_2 \geq \cdots \geq \eta_k \geq \lambda_k \geq \cdots \geq \eta_{k+p} \geq \eta_{k+p+1} = 0 \text{ and } \sum \eta_i = \sum \lambda_i + p \tag{2}$$

Proof. Clearly, for all $i \in [k]$, $\mu_i \geq \lambda_i$, since for each row, either boxes are added or not. Suppose for the sake of contradiction that $\mu_{j+1} > \lambda_j$ for some $j \in [k]$. So $\mu_j \geq \mu_{j+1} > \lambda_j \geq \lambda_{j+1}$. But then a box was added to the $(\lambda_j + 1)$-column of both row j and row $j+1$, a contradiction. Next, since no two boxes are added to the same column, a maximum of one one new row can be added and so $\mu_{k+1} \geq \mu_{k+2} = 0$. Finally, at the end of adding p boxes, of course the total number of boxes exactly increased by p. So of course $\sum \mu_i = \sum \lambda_i + p$ and we see that (1) holds.

Now, at most one box is added to each row of η so of course either $\eta_i = \lambda_i$ or $\eta_i = \lambda_i + 1 \implies \eta_i - \lambda_i \in \{0, 1\}, \forall i \in [k]$. So then if the result is still a partition, which is assumed by the idea that it has a Schur Polynomial, of course $\eta_1 \geq \eta_2 \geq \cdots \eta_k \geq \eta_{k+1}\eta \cdots \geq \eta_{k+p-1} \geq \eta_{k+p} \geq \eta_{k+p+1} = 0$. So then since $\lambda_i \geq \lambda_{i+1}$, $\eta_{i+1} = \lambda_{i+1} + 1$ and $\eta_i \leq \eta_{i+1} \implies \eta_i \geq \lambda_i \geq \eta_{i+1},\ \forall i \in [k]$. Then, since at most k boxes are added to rows 1 through k, 0 to p new rows may be added and so $\eta_{k+1}, \ldots, \eta_{p+k} \in \{0, 1\}$ and we may now even add more than p boxes so the maximum number of parts will be $p + k$. Finally, at the end of adding p boxes, of course the total number of boxes exactly increased by p. So of course $\sum \eta_i = \sum \lambda_i + p$ and (2) holds.

□

Problem 2.2. For partitions λ and μ of the same integer, show that in fact $K_{\lambda\mu} \neq 0$ if and only if $\mu \trianglelefteq \lambda$.

Proof. Let $\lambda = (\lambda_1, \ldots, \lambda_n)$ and $\mu = (\mu_1, \ldots, \mu_m)$ partition the same integer. Recall that $K_{\lambda\mu}$ counts SSYT of shape λ and content μ. So then trivially $K_{\lambda\mu} \neq 0$ if and only if there even exists an SSYT of shape λ having exactly μ_i entries equal to i for each $i \in [m]$.

($\implies$) Suppose that $K_{\lambda\mu} \neq 0$ and that T is such a SSYT. Since entries strictly increase down columns, every entry in row k is at least k. That is, iteratively we can see that in row 1, everything is at least 1, then since row 2 must be strictly larger component-wise, everything is at least $2, \ldots,$ and so on; every entry equal to each $k \in [m]$ appears before or at but not after row k of T. Therefore all entries equal to $1, 2, \ldots, j$ must lie in the first j rows of T.

Now, the first j rows of T contain exactly $\lambda_1 + \lambda_2 + \cdots + \lambda_j$ boxes and all entries $1, \ldots, j$ must occur in those rows. So then we must have as many entries equal to each $k \in [j]$ as there are boxes on row 1 through j. That is,

$$\mu_1 + \mu_2 + \cdots + \mu_j \leq \lambda_1 + \lambda_2 + \cdots + \lambda_j$$

for every $j \in [n]$. So then by definition $\mu \trianglelefteq \lambda$.

($\impliedby$) On the other hand, if $\mu \trianglelefteq \lambda$, we construct such a SSYT of shape λ and content μ.

Fill the boxes of λ from left to right, top row to bottom row, by placing first μ_1 copies of 1, then μ_2 copies of 2, and so on for each $i \in [m]$, which can be done at each step $j \to j+1 \in [m]$ since we must have at least as many boxes $\lambda_1 + \cdots + \lambda_k$ in rows 1 through k as there are copies of 1 through k; $\mu_1 + \mu_2 + \cdots + \mu_j \leq \lambda_1 + \lambda_2 + \cdots + \lambda_j$. Finally, we have that μ, λ partition the same integer. So once we run out of copies of integers in $[m]$ encoded by μ to fill λ in with, we must have also run out of boxes in λ to put things into. So we have constructed a tableau of shape λ with content $\mu \implies K_{\lambda\mu} \neq 0$.

Thus,

$$K_{\lambda\mu} \neq 0 \quad \iff \quad \mu \trianglelefteq \lambda.$$

$\square$

Problem 3.1. If w is Knuth equivalent to the word of a tableau T, show that the number of rows of T is the longest strictly decreasing sequence that can be extracted from w. Show that the total number of boxes in the first k columns of T is the maximum sum of the lengths of k disjoint strictly decreasing sequences that can be extracted from w.

Proof. Let λ be the shape of T.

By **Lemma 1**, if w_T is the word of T, then $L(w_T, k) = \lambda_1 + \cdots + \lambda_k$ and by **Lemma 2** since w is knuth equivalent to w_T, we have that

$$L(w_T, k) = \lambda_1 + \cdots + \lambda_k = L(w, k).$$

Now we apply the same result to decreasing subsequences. Consider the conjugate shape λ'. Analagously, the maximum total length of k disjoint strictly decreasing subsequences extracted from w_T and therefore w must then be

$$\lambda'_1 + \cdots + \lambda'_k.$$

So for $k = 1$, the longest strictly decreasing subsequence has length λ'_1 , which is exactly the number of rows of T. More generally, λ'_i is the number of boxes in the i-th column of T and then $\lambda'_1 + \cdots + \lambda'_k$ is exactly the total number of boxes in the first k columns of T. So we see that the total number of boxes in the first k columns of T is fact exactly the maximum sum of the lengths of k disjoint strictly decreasing subsequences extracted from w.

$\square$

Problem 3.2. Deduce from this a result of Erdos and Szekeres: Any word of length greater than n^2 must contain either an increasing or a strictly decreasing sequence of length greater than n. Show that this is sharp. More generally, a word of length greater than $m \cdot n$ must contain an increasing sequence of length greater than n or a decreasing sequence of length greater than m.

Proof. Let w be a word, and let T be its insertion tableau which has shape λ. Suppose w has no increasing subsequence of length greater than n, and no strictly decreasing subsequence of length greater than m.

At the beginning of the chapter $L(w, 1)$ is defined as the longest increasing subsequence. By our Lemmas, this is exactly $L(w, 1) = \lambda_1$, the number of boxes in the first row of T. So by our assumption the number of boxes in row 1 of T is at most n; $\lambda_1 \leq n$ We proved in the previous problem that the longest strictly decreasing subsequence of $w_T = w$ here has length m equal to the number of rows of T. So then λ must have at most m parts.

Putting everything together, the length of w is equal to the number of boxes in T, which must have at most n boxes in row 1 and therefore all of its rows and also there are at most m rows in T. That is,

$$|w| = |T| \leq mn.$$

Therefore, contrapositively, if w has length greater than or equal to mn, then w either contains an increasing subsequence of length greater than n, or a strictly decreasing subsequence of length greater than m. Setting $m = n$, we see that any word of length greater than n^2 contains either an increasing subsequence or a strictly decreasing subsequence of length greater than n.

Now we show sharpness for the n^2 case. To show sharpness in the case $m = n$, consider the word

$$\omega = \underbrace{n\,n\,\cdots\,n}_{n \text{ times}}\underbrace{(n-1)\,(n-1)\,\cdots\,(n-1)}_{n \text{ times}}\cdots\underbrace{1\,1\,\cdots\,1}_{n \text{ times}}.$$

ω has length n^2 and obviously the longest increasing subsequence is $nn \cdots n$ which has length n and the longest strictly decreasing subsequence that appears is $n(n-1)\cdots 21$ which has length n. No larger examples exist for either types of subsequences of ω. So the bound is strict for n^2; we can only guarantee in full generality that there appears a **length n or greater** increasing or strictly decreasing sequence, claiming a higher bound would yield the counterexample ω we just constructed.

□

Problem 3.3. Find a word of length 6 with $L(w,1)=4$, $L(w,2)=6$, but which does not have two disjoint increasing subsequences of lengths 4 and 2.

Proof. Let $w=(236145)$. The longest increasing subsequence is (2345) and no other length $L(w,1)=4$ increasing sequence exists. $w=(236)\sqcup(145)$ a disjoint union of two increasing subsequence, so the longest disjoint union is the length of the whole word; $K(w,2)=6$. Lastly since the only subsequence of length 4 is (2345), the only subsequence of length 2 disjoint from it is the remainder of the word (61), which is not increasing. So $w=(236145)$ satisfies all conditions.

$\square$

Problem 4.6. Show that the number of involutions in S_n is

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!(2^k k)!}.$$

Proof. An involution in S_n is a permutation which is it's own inverse. That is, it is a product of disjoint transpositions and 1-cycles. So then an involution is constructed by forming $k \in [n]$ disjoint transpositions and fixing the remaining $n-2k$ elements of $[n]$. We describe this construction step by step below to count these involutions.

For any $k \in [n]$. There are $\binom{n}{2k}$ ways to select the $2k$ elements that will appear in transpositions. Then the number of ways to partition these $2k$ elements into pairs is the iterative product below, where we divide by $k!$ since order we choose the pairs doesn't matter here:

$$\begin{aligned}\frac{\prod_{j=0}^{k-1} \binom{2k-2j}{2}}{k!} &= \frac{1}{k!} \prod_{j=0}^{k-1} \frac{(2k-2j)!}{2!((2k-2j)-2)!} = \frac{1}{k!} \prod_{j=0}^{k-1} \frac{(2k-2j)((2k-2j)-1)}{(2)} \\ &= \frac{(2k(2k-1))((2k-2)(2k-3)) \cdots ((2)(1))}{2^k k!} = \frac{(2k)!}{2^k k!}.\end{aligned}$$

Combining the two we get that the number of involutions with exactly $k \in [n]$ transpositions is

$$\binom{n}{2k} \frac{(2k)!}{2^k k!} = \frac{n!}{(n-2)!} \frac{1}{2^k k!} = \frac{n!}{(n-2k)! 2^k k!}.$$

We can form between $k = 1$ and $k = \lfloor n/2 \rfloor$ transpositions total from $[n]$. So finally we have that the number of involutions in S_n is

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)! 2^k k!}.$$

$\square$

Problem 6.1. Prove the following identities

$$ne_n(x) - p_1(x)e_{n-1}(x) + p_2e_{n-2}(x) - \ldots + (-1)^n p_n(x) = 0 \tag{3}$$
$$nh_n(x) - p_1(x)h_{n-1}(x) + p_2h_{n-2}(x) - \ldots - p_n(x) = 0 \tag{4}$$

Proof. Recall

$$E(t) = \sum_{r\geq 0} e_r(x)t^r = \prod_i (1 + x_i t), \qquad H(t) = \sum_{r\geq 0} h_r(x)t^r = \prod_i \frac{1}{1 - x_i t}.$$

We begin by observing that $E'(t) = \sum_{r\geq 1} re_r(x)t^{r-1}$. Also,

$$\frac{E'(t)}{E(t)} = \sum_i \frac{x_i}{1 + x_i t} = \sum_i x_i(1 - x_i t + x_i^2 t^2 - \cdots) \text{ (By Geometric series)}$$
$$= p_1(x) - p_2(x)t + p_3(x)t^2 - \cdots = \sum_{r\geq 1} (-1)^{r-1} p_r(x)t^{r-1}.$$
$$E'(t) = E(t) \sum_{r\geq 1} (-1)^{r-1} p_r(x)t^{r-1} \implies \sum_{r\geq 1} re_r(x)t^{r-1} = \left(\sum_{q\geq 0} e_q(x)t^q\right)\left(\sum_{r\geq 1} (-1)^{r-1} p_r(x)t^{r-1}\right).$$

Thus,

$$[t^{n-1}]E'(t) = ne_n(x) = p_1(x)e_{n-1}(x) - p_2(x)e_{n-2}(x) + \cdots + (-1)^{n-1}p_n(x)$$
$$\implies ne_n(x) - p_1(x)e_{n-1}(x) + p_2(x)e_{n-2}(x) - \cdots + (-1)^n p_n(x) = 0.$$

Similarly, we observe that $H'(t) = \sum_{r\geq 1} rh_r(x)t^{r-1}$. Also,

$$\frac{H'(t)}{H(t)} = \sum_i \frac{x_i}{1 - x_i t} = \sum_i x_i(1 + x_i t + x_i^2 t^2 + \cdots) \text{ (By Geometric series)}$$
$$= p_1(x) + p_2(x)t + p_3(x)t^2 + \cdots = \sum_{r\geq 1} p_r(x)t^{r-1}.$$
$$H'(t) = H(t) \sum_{r\geq 1} p_r(x)t^{r-1} \implies \sum_{r\geq 1} rh_r(x)t^{r-1} = \left(\sum_{q\geq 0} h_q(x)t^q\right)\left(\sum_{r\geq 1} p_r(x)t^{r-1}\right).$$

Thus,

$$[t^{n-1}]H'(t) = nh_n(x) = p_1(x)h_{n-1}(x) + p_2(x)h_{n-2}(x) + \cdots + p_n(x)$$
$$\implies nh_n(x) - p_1(x)h_{n-1}(x) - p_2(x)h_{n-2}(x) - \cdots - p_n(x) = 0.$$

□

Problem 6.6. Show that $s_\lambda(1,\dots,1) = \prod_{i<j} \frac{\lambda_i-\lambda_j+j-i}{j-i}$

Proof. By Exercise 5,

$$s_\lambda(1,x,x^2,\dots,x^{m-1}) = x^r \prod_{i<j} \frac{x^{\lambda_i-\lambda_j+j-i}-1}{x^{j-i}-1},$$

where

$$r = \lambda_2 + 2\lambda_3 + \cdots = \sum_i (i-1)\lambda_i.$$

Using the well-known identity, we get the following:

$$x^a - 1 = (x-1)(1+x+\cdots+x^{a-1}) \implies \frac{x^{\lambda_i-\lambda_j+j-i}-1}{x^{j-i}-1} = \frac{1+x+\cdots+x^{\lambda_i-\lambda_j+j-i-1}}{1+x+\cdots+x^{j-i-1}}.$$

Therefore

$$s_\lambda(1,x,x^2,\dots,x^{m-1}) = x^r \prod_{i<j} \frac{1+x+\cdots+x^{\lambda_i-\lambda_j+j-i-1}}{1+x+\cdots+x^{j-i-1}}.$$

Now set $x=1$. Since $x^r = 1$, and

$$a = 1 + (1) + \cdots + (1)^{a-1}$$

we get that

$$s_\lambda(1,\dots,1) = \prod_{i<j} \frac{\lambda_i-\lambda_j+j-i}{j-i}.$$

As desired.

$\square$